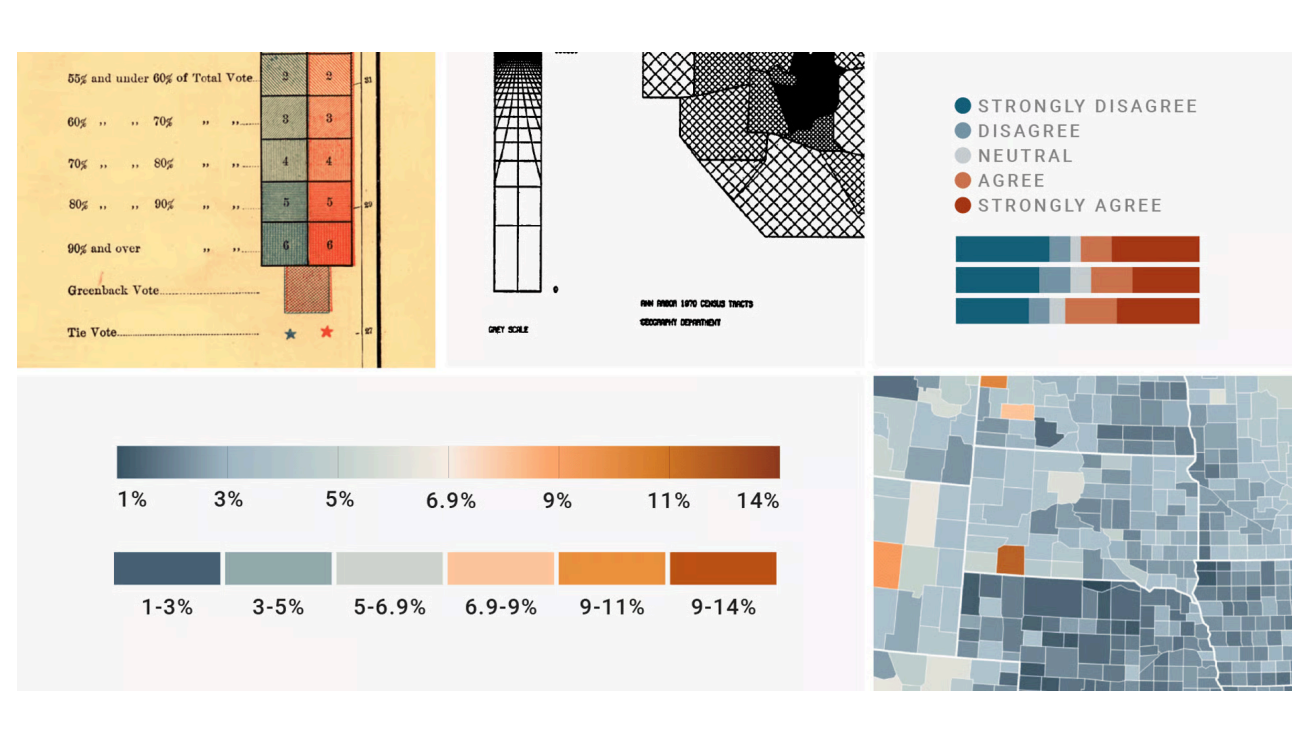
Color in data vis
12 min
March 16th, 2021

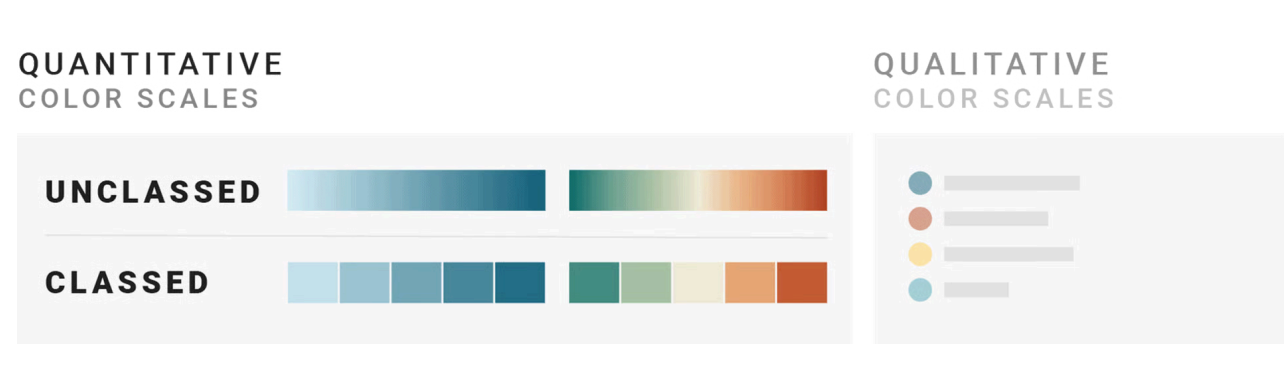
When to use classed and when to use unclassed color scales

Lisa Charlotte Muth



This is part 4 of a series on "Which color scale to use when visualizing data". Part 1 / Part 2 / Part 3

If you use a quantitative color scale, you need to decide if you want to split your values into brackets (make it *classed*, also called *classified*, *stepped*, *quantized*, *graduated*, *binned* or *discrete*) or not (leave it *unclassed*, also called *continuous*). This article will help you to make that decision.

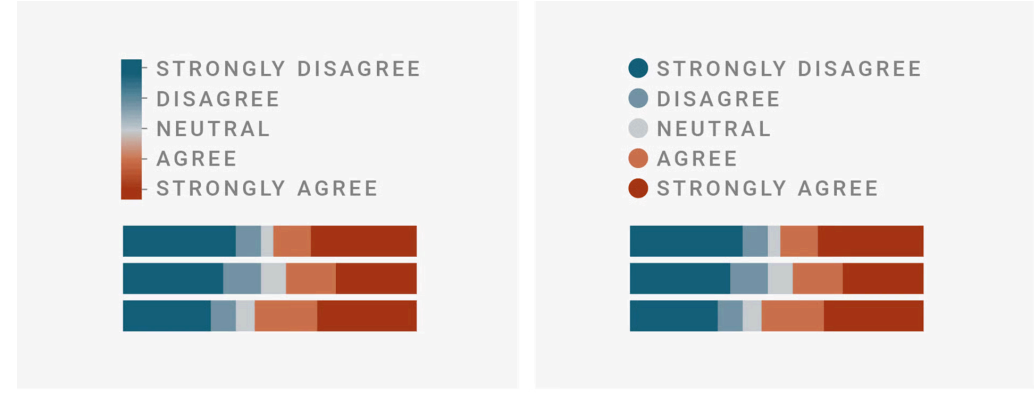


Index

1 Use a classed color scale if the data is classed
A bit of history: Suddenly, it's possible. But should we do it?
2 Use a classed color scale if you want to communicate statistical brackets
3 Use an unclassed color scale to give a nuanced view
4 Use an unclassed color scale if you don't want to interpret for the reader
5 Use a classed color scale if you want readers to read values
A bit of context: Maybe it's not that important

1 Use a classed color scale if the data is classed

First things first: If your data isn't continuous, don't use a continuous color scale. Meaning, when visualizing ordinal data like Likert scales (strongly disagree, disagree, neutral, agree, strongly agree), clothing sizes (XS, S, M, L, XL), official ranks, etc., use a classed color scale. An unclassed color scale would suggest that there are options in between when there are not.



NOT IDEAL

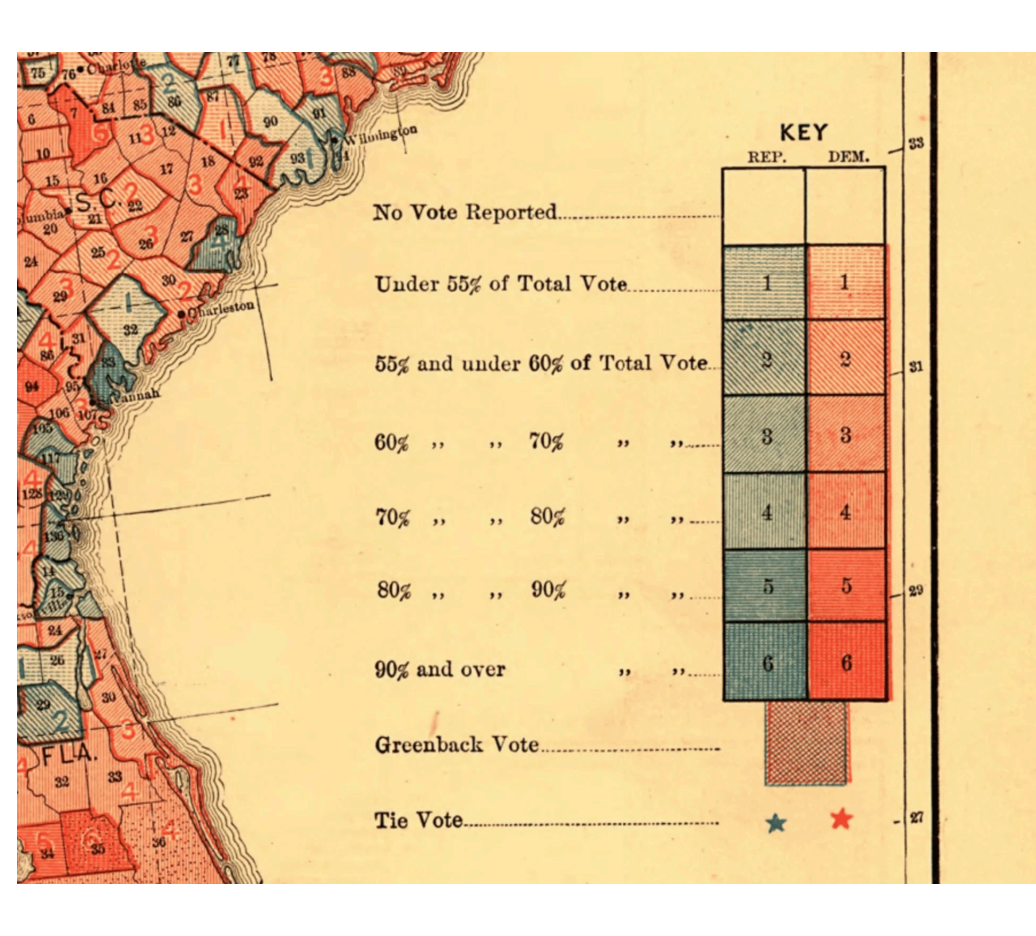
BETTER

But every continuous data *can* be made discrete: You can always sort unemployment rates into ranges (like 2-3%, 3-4%, 4-5%, etc.) and give each range a color instead of assigning each value to one color on a gradient.

So when *should* you class your color scale?

A bit of history: Suddenly, it's possible. But should we do it?

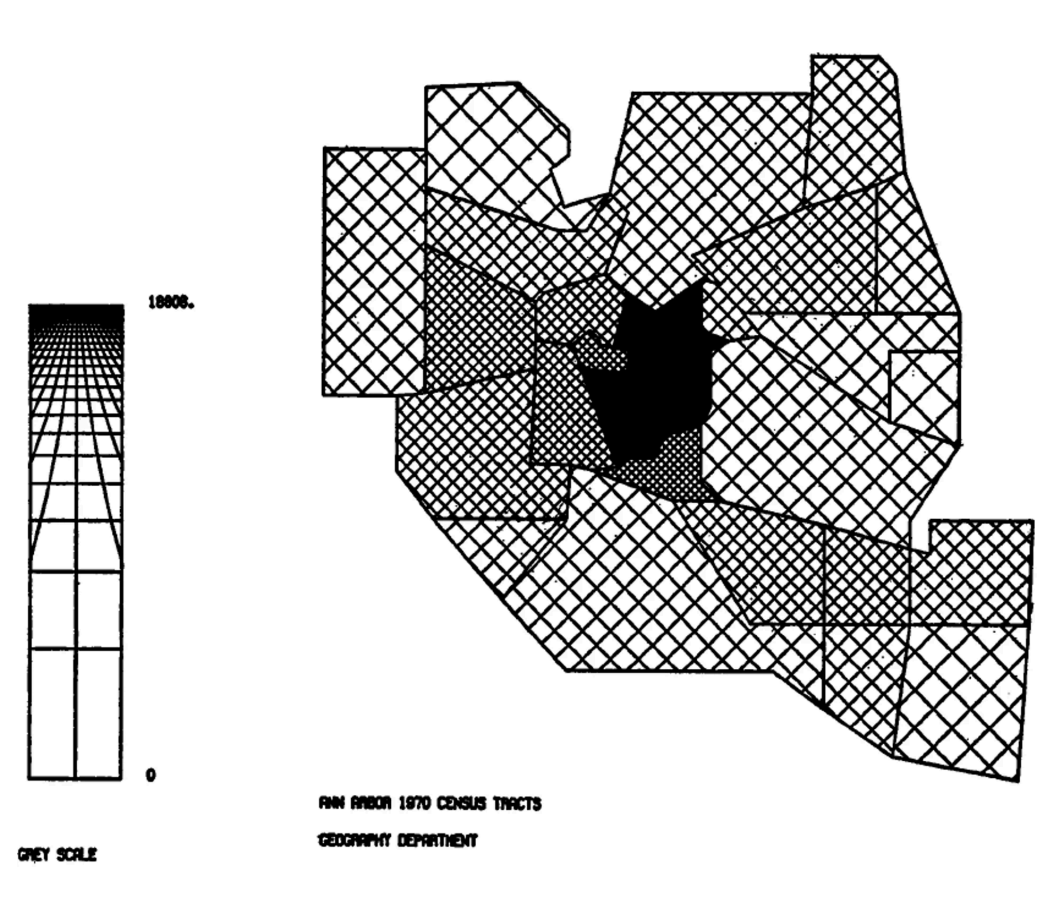
For a long time, cartographers and data visualization designers didn't have to ask that question. They had to use classed color scales – because it proved to be infinitely difficult to calculate, paint and print unclassed ones. Here's a map from 1883:



A classed map showing the popular vote in the US Election 1880. From Scribner's statistical atlas of the United States, 1883. Full map from The Library of Congress.

The diverging color scale in this map uses two colors: blue and red. The creator(s) printed these colors thinly to create light colors and used dense hatching for darker colors. That was what the technology at the time allowed for.

In a short research note in July 1973, cartographer Waldo R. Tobler used hatching, too, to create what was probably the first unclassed map. "It is now technologically feasible to produce virtually continuous shades of grey by using automatic map drawing equipment," he wrote[1], and showed the following map:



"Choropleth Map Without Class Intervals", 1973, by W. R. Tobler.

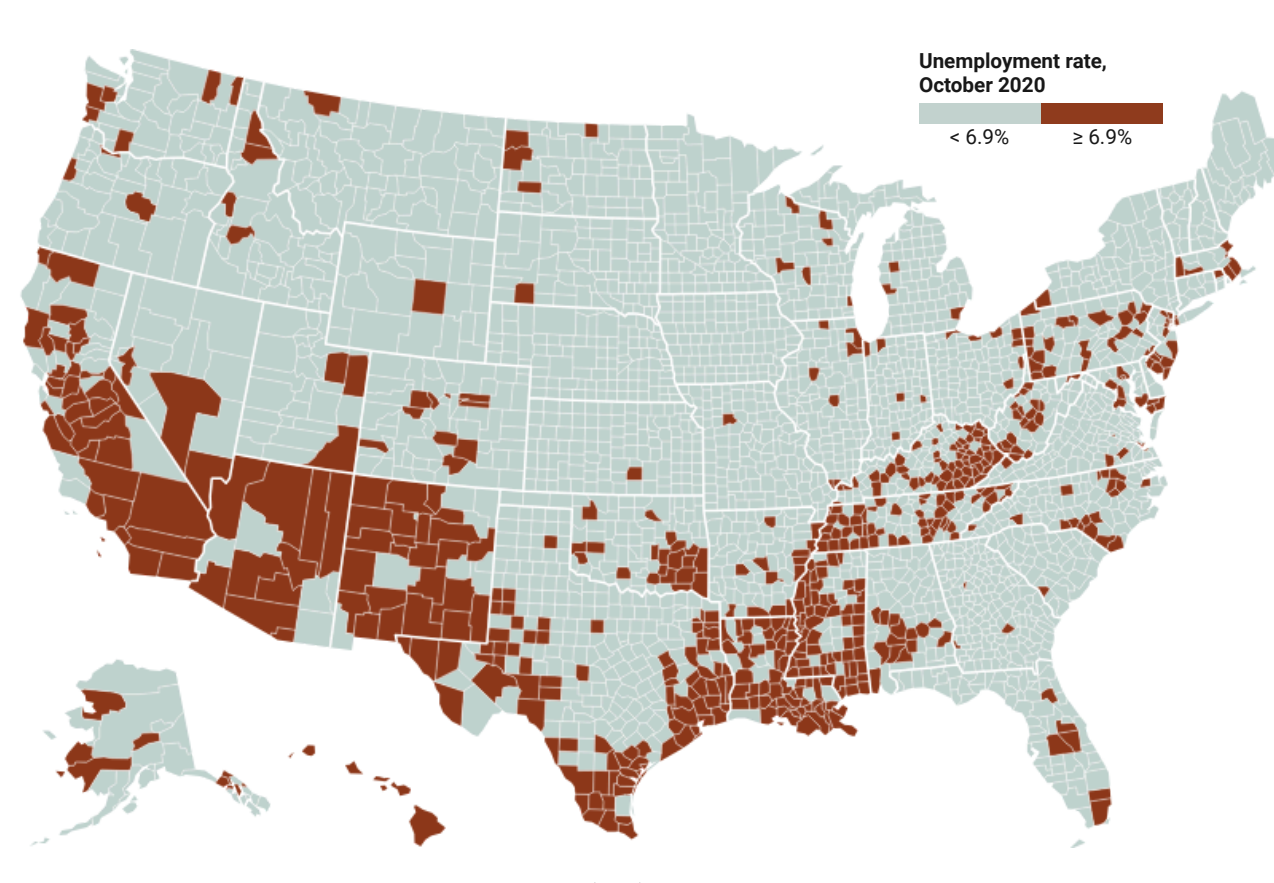
Not classing numbers anymore? That was a provocative idea in the 1980s and 1990s. Cartographers asked: "Should we create these new unclassed maps that go against everything we did in the past hundreds of years; against everything we're used to?" (Not an actual quote.) Studies were written; comments were published.

Here's what they found out.

2 Use a classed color scale if you want to communicate statistical brackets

With classed color scales, you can make a point more easily than with unclassed color scales. You can group values (and therefore regions) with the same color together so your readers can quickly see the statement you want to make.

Cartographer Michael Dobson was a big advocate for classed maps in the 1980s. He called it a "simpler, more efficient communication device."[2] It definitely is simpler – simplifying is the whole point of a classed map. But is it also more efficient? That depends on what you want to communicate.



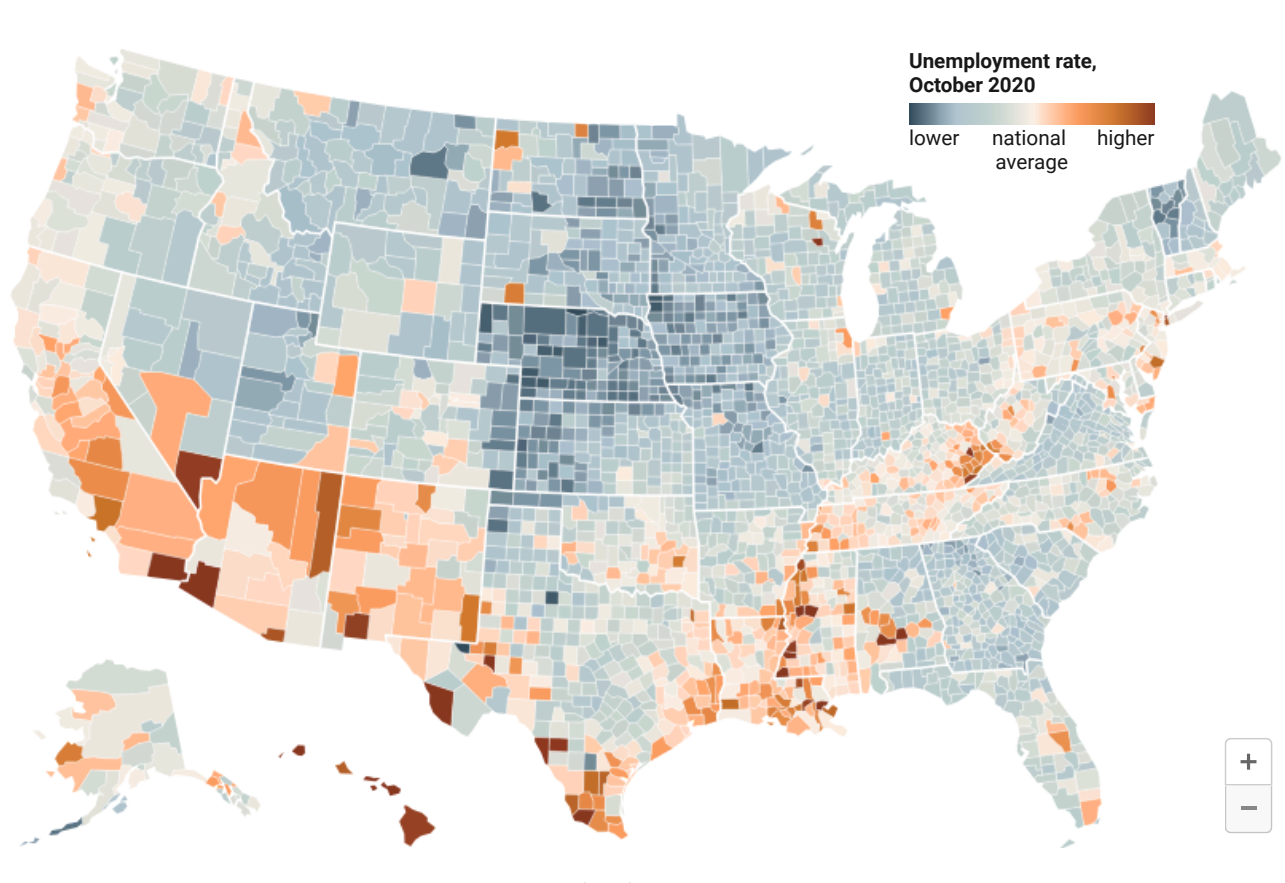
Counties with a unemployment rate above the national average (6.9%) and below in October 2020
Source: BLS/LAUS • Get the data • Created with Datawrapper

If you want to communicate something very narrow, classed maps are a great choice. Cartographers Katherine Mak and Michael R.C. Coulson pointed out in 1991[3] that **as soon as a map has a statistical objective**, like showing

- "the lowest one-tenth of counties by medium family income,"
- "areas with cancer mortality rates more than two standard deviations above the mean"
- if a county has a higher unemployment rate than the national average, like in the map above

Classes are the way to achieve that. "**The classification system defines itself** and the focus is on which data units fall into a particular predefined class, rather than the overall pattern of the map." If you want your readers to see if certain regions fall into a **statistical bracket**, use classes.

But as soon as you want to show a **general pattern** – e.g., "temperatures are higher in the south than in the north" or "our revenue was higher this year than last year" – an unclassed map is probably the better choice. Here's the same unemployment map with a diverging color scale:



Counties with a unemployment rate above the national average (6.9%) and below in October 2020
Source: BLS/LAUS • Get the data • Created with Datawrapper

Note how this map makes it harder to see which statistical brackets the counties fall in – meaning, if they have an unemployment rate below or above the national one.

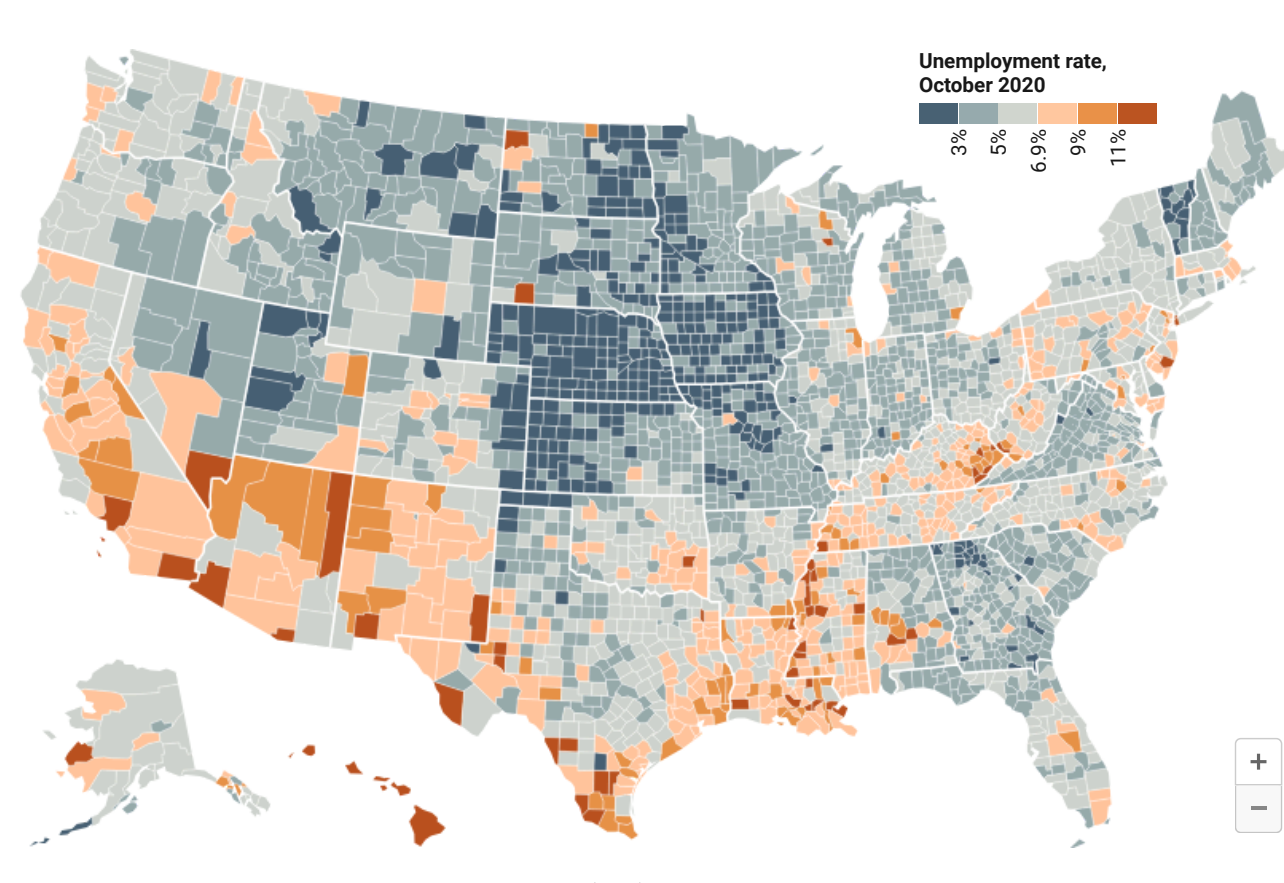
3 Use an unclassed color scale to give a nuanced view

As the two maps above make clear, classed maps show less nuance than unclassed maps.

The unclassed map gives a more truthful, nuanced view of the unemployment rate. **"The unclassed choropleth is the most exact representation of the data model possible,"** Judith A. Tyner writes in her "Principles of Map Design"[4]

Communicating the complexity of the data is a noble goal in itself. If this is high on your priority list, go for unclassed maps.

But classed maps, too, can be more or less nuanced. **The more classes you show, the more nuanced your map becomes**[5]. The unemployment rate map showing only two classes is an extreme example. So here's the same map with a six classes, three above and three below the national average:



Counties with a unemployment rate above the national average (6.9%) and below in October 2020
Source: BLS/LAUS • Get the data • Created with Datawrapper

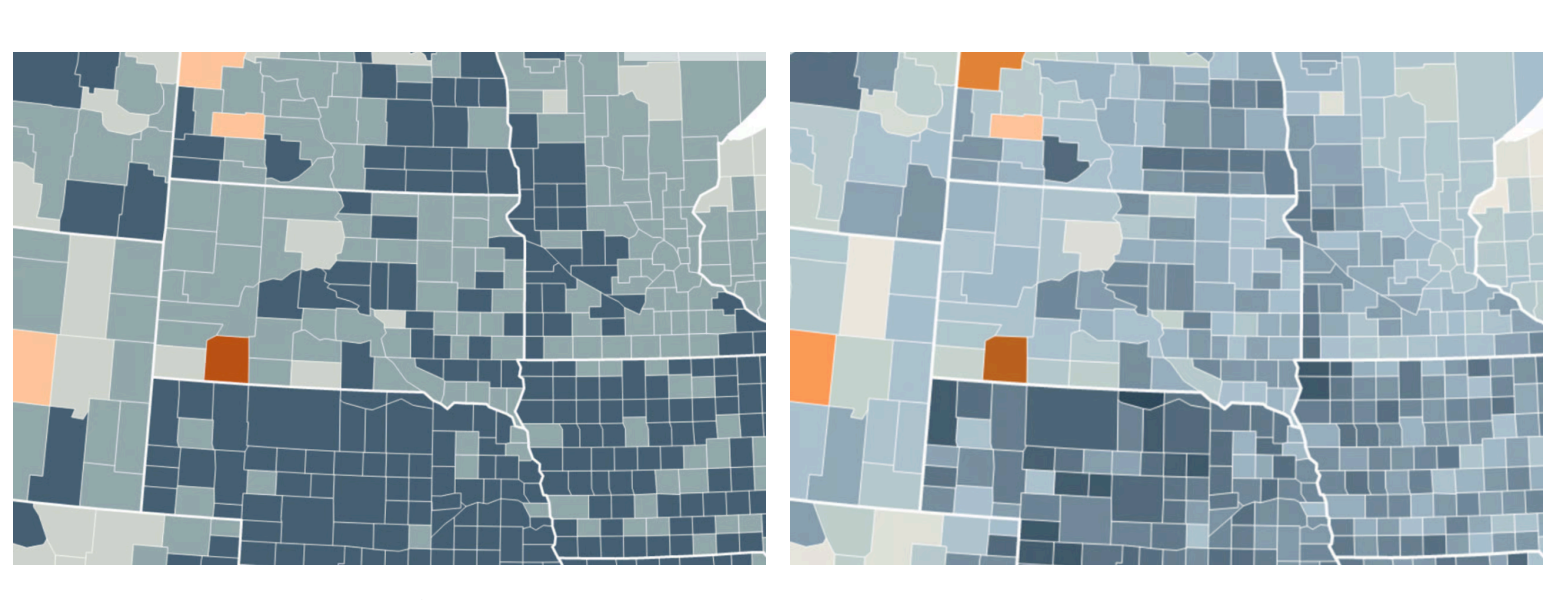
This map shows more nuance than the two-class map – but less nuance than the unclassed map.

4 Use an unclassed color scale if you don't want to interpret for the reader

The subtle differences you can see in an unclassed map enable readers to see things that are not important for you – but are for them. Unclassed maps make it easier to see…

- …**outliers**. In classed maps, they're often placed together in a bracket with other regions with high or low numbers.
- **…if the transition between clusters is smooth or abrupt.**
- …if the region I'm interested in as a reader has a higher or lower value than the **neighboring regions**.

To show an example, let's zoom in on South Dakota – the state in the middle of the following images. South Dakota has an unemployment rate closer to the national average (making its blue lighter) than most surrounding states. On the unclassed map, we can see that almost all South Dakota counties that border other states have a higher unemployment rate than the counties on the other side of the state border.



South Dakota and surrounding states. Left: map with six classes. Right: unclassed map.

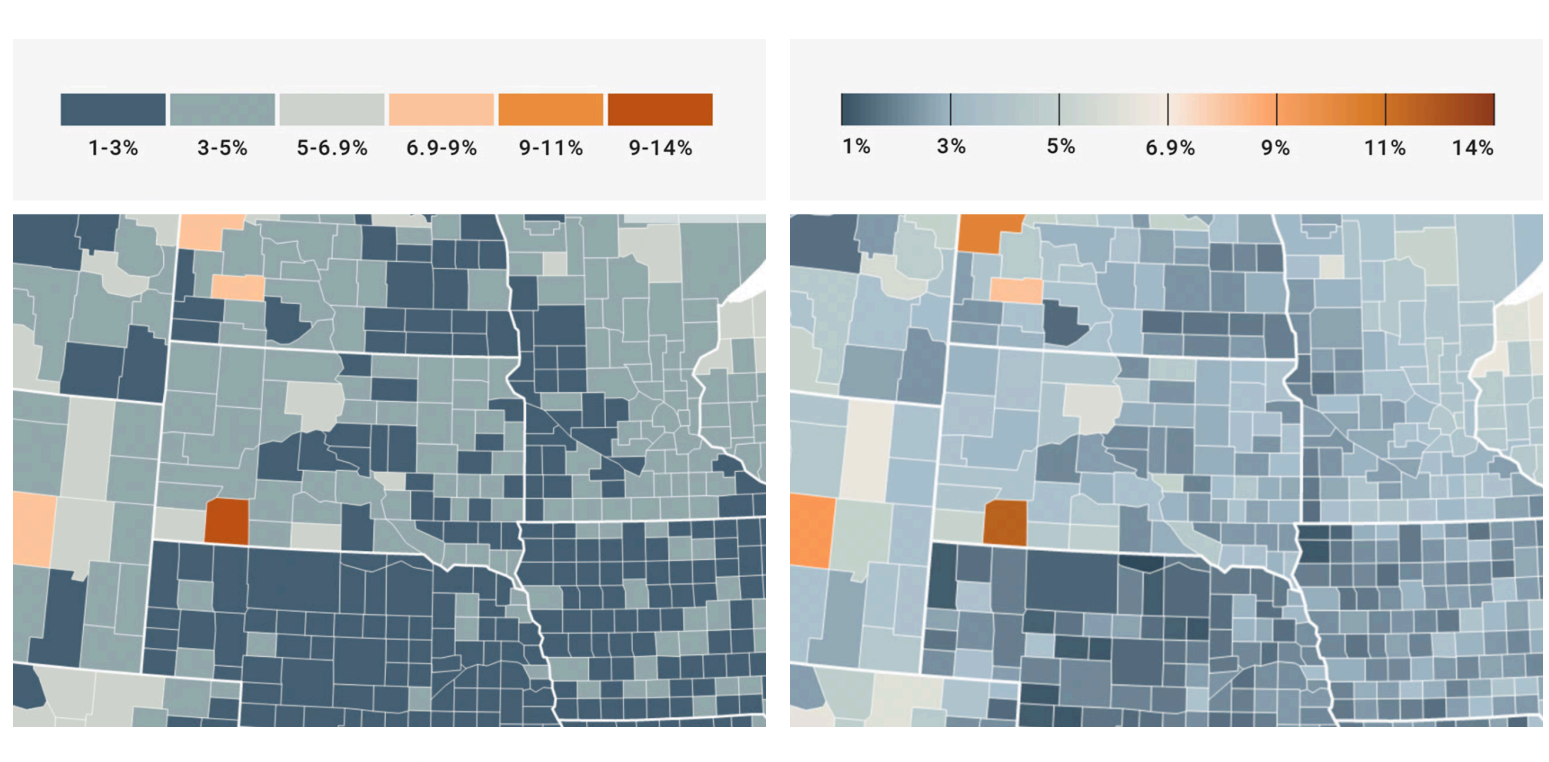
In the classed map, these subtle differences are not visible. Readers can't know if the dark blue counties in South Dakota have a slightly higher or lower unemployment rate than the dark blue counties in all the surrounding states.

If I lived in South Dakota, I'd rather see the unclassed map. **"How does my region compare?" is a question that's better answered with an unclassed map.**

5 Use a classed color scales if you want readers to read values

Classed maps enable readers to read only ranges (e.g., between 6% and 7%) – but these ones, they can read well. Mak and Coulson, who we already met a few sections ago, concluded after a study in 1991 that "classed maps have a distinct and statistical significant advantage over unclassed maps in value-estimation tasks."[6]

Let's have a look at South Dakota again:



South Dakota and surrounding states. Left: map with six classes. Right: unclassed map.

On the classed map above, you can be sure that you read the value range for the regions correctly – while on the unclassed map, you can only have "good guesses". They might be more narrow ("close to 6%" instead of "between 5% and 6.9%"), but they remain guesses.

But value readability, too, depends on the number of classes: The more classes you have, the less likely it becomes that readers will read the values correctly.

So if you want readers to read specific value ranges, go for a classed map with only a few classes. That's especially important **when you show static maps** (e.g., in print or PDF reports) where people don't have the option to see tooltips or hover over a legend.

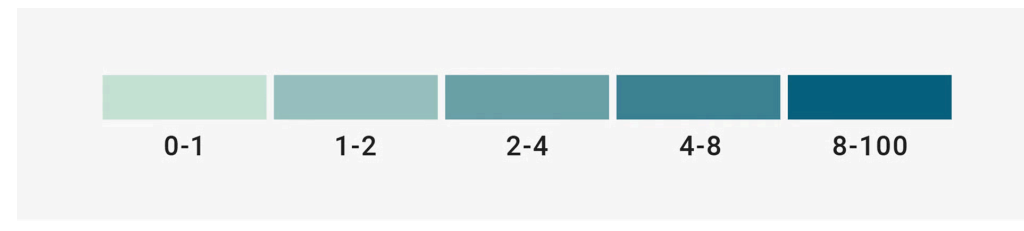
A bit of context: Maybe it's not that important

Showing nuance, enabling readers to read values – all this not only depends on the decision if you sort your values into classes, but also from the interpolation, the number of classes, if you chose a sequential or diverging color palette, the hues, the legend you choose, etc.

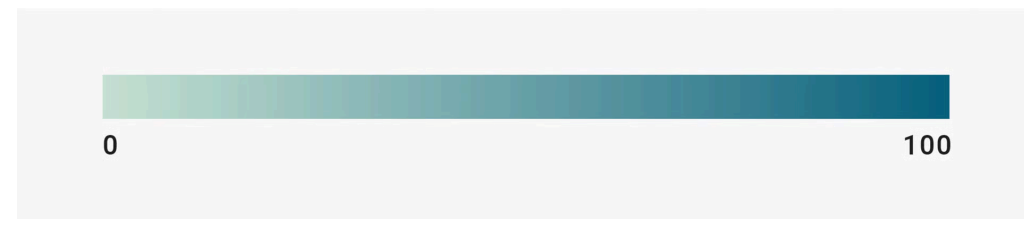
So when Boren Dent writes in his classic "Cartography – Thematic Map Design" that "the burden of deciding on a classification scheme is removed from the designer" when choosing unclassed maps[7], I'd argue that this is a small burden to lift, considering there are so many challenges remaining.

And when studies and comments from the 1990s use the argument "interpretation is left to the reader" against unclassed maps, I'd point out that all these tools I just listed can still steer interpretations in very different directions.

I write about the complications of finding a good interpolation here, so let's take legends as an example. Which one will enable readers to correctly read values in a map – the unclassed one or the classed one?



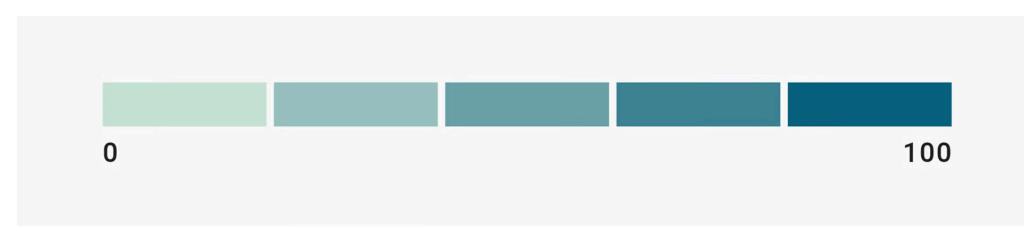
Legend for a classed color scale



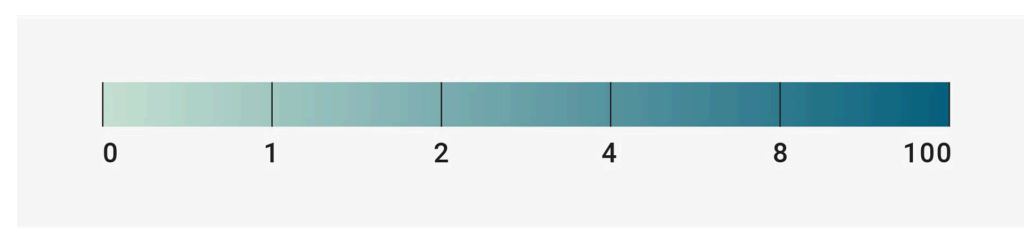
Legend for an unclassed color scale

Most people (including you, I hope) will agree that the classed legend makes it easier.

How about now?



Legend for a classed color scale



Legend for an unclassed color scale

"Well, duh, the unclassed one is way better now," I hear you say.

Creating quantitative color scales comes with a lot of challenges. Creating a good legend is one of them. Deciding for a classed or unclassed color scale is another – one that, hopefully, got a bit easier after reading this article.

No matter if you decide for a classed map at the end, you should **start your process by looking at an unclassed map.** This will help you see subtle differences between regions and make a conscious decision if and how you should simplify them. (That's why Datawrapper choropleth maps use a continuous color scale by default.)

This was the last part of a series on "Which color scale to use when visualizing data" (Part 1 / Part 2 / Part 3). Do you know of any other reasons to use classed instead of unclassed color scales – or the other way round? Please leave them in the comments or let me know at lisa@datawrapper.de. Thanks for reading!

References

- Katherine Mak and Michael R.C. Coulson: **Map-User Response to Computer-Generated Choropleth Maps**: Comparative Experiments in Classification and Symbolization, 1991, p110.
- Robert G. Cromley: **Classed versus Unclassed Choropleth Maps**. A Question of How Many Classes. In: Cartographica: The International Journal for Geographic Information and Geovisualization, Volume 32, issue 4, 1995.
- Boren D. Dent, Jeffrey S. Torguson, Thomas W. Hodler: **Cartography. Thematic Map Design**, sixth Edition, 2009.
- Judith A. Tyner: **Priciples of Map Design**, 2010.

[1] W. R. Tobler: Choropleth Maps Without Class Intervals?, p262. In: Geographical Analysis, Volume 5, Issue 3, July 1973. ↩

[2] Michael W. Dobson: Commentary: Perception of Continuously Shaded Maps. In: Annals of the Association of American Geographers 70, 1980, p107. ↩

[3] Katherine Mak and Michael R.C. Coulson: Map-User Response to Computer-Generated Choropleth Maps: Comparative Experiments in Classification and Symbolization, 1991, p110. Emphasis added. ↩

[4] Judith A. Tyner: Priciples of Map Design, 2010, p164. ↩

[5] Robert G. Cromley makes this point in his paper Classed versus Unclassed Choropleth Maps. A Question of How Many Classes: "Although the debate about classed versus unclassed maps has focused on this choice as a dichotomy, the choice can be expressed as a choice to determine the proper number of classes needed to represent the data distribution." In: Cartographica: The International Journal for Geographic Information and Geovisualization, Volume 32, issue 4, 1995, p19 ↩

[6] Katherine Mak and Michael R.C. Coulson: Map-User Response to Computer-Generated Choropleth Maps: Comparative Experiments in Classification and Symbolization, 1991, p120. ↩

[7] Boren D. Dent, Jeffrey S. Torguson, Thomas W. Hodler: Cartography. Thematic Map Design. Sixth Edition, 2009, p109. ↩

Lisa Charlotte Muth (she/her, @lisacmuth, @lisacmuth@vis.social) is Datawrapper's head of communications. She writes about best practices in data visualization and thinks of new ways to excite you about charts and maps. Lisa lives in Berlin.

Liked this article? Maybe your friends will too:
LinkedIn Bluesky

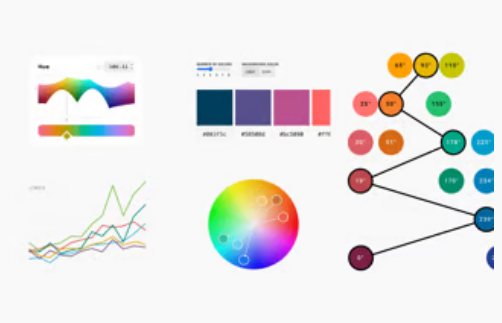
COMMENTS

To post a comment you need to sign in via
Twitter or Github or Mastodon

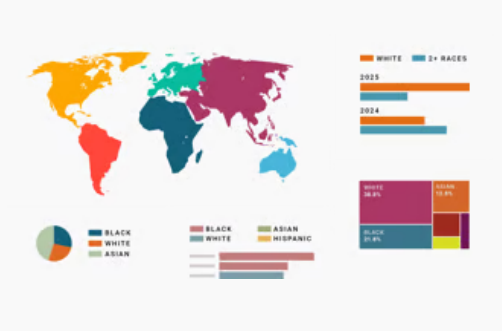
NEWSLETTER

Sign up to our newsletters to get notified about everything new on our blog.

RELATED POSTS



How to find & create good color palettes
December 11th, 2024 by Lisa Charlotte Muth



What to consider when choosing colors for race, ethnicity, and world regions
October 9th, 2024 by Lisa Charlotte Muth

Remind readers of the colors in your data visualization
October 11th, 2023 by Lisa Charlotte Muth

ALL BLOG CATEGORIES
Climate vis
Covid vis
Data vis do's & don'ts
How to deal with data
Opinions
Under the hood
Weekly Charts
Color book updates
Data Vis Book Club
Datawrapper News
Map Stories
Team news and hiring
Unwrapped
Color in data vis
Data Vis Dispatch
Fix my chart
Maps
Uncategorized
User stories
View a chronological list of our articles
Learn more about our newsletters

Product
Charts
Maps
Tables
Pricing

Feature highlights
Custom themes
Dark mode
Print export
Localization
Accessibility
Privacy
Security

Solutions
Media
Finance
Government
Case studies

Resources
Blog
Academy
FAQ
River
Webinars
Training materials
API Docs
Get support

About
About us
Latest improvements
Careers
Privacy policy
Terms of service
Imprint
Contact us

© 2025 Datawrapper is developed by Datawrapper GmbH.